

LUCIDPREP

Grade 7 Mathematics

U.S. Standards-Aligned Course Syllabus

MATH 7

Math 7 · Common Core

36 weeks • 4 modules • 70% mastery

Curriculum Architecture

This course is organized as 36 weeks across four 9-week modules. Every week follows LucidPrep's established secondary learning cycle: AI Instructor opening, objectives, vocabulary, five warm-ups, main lecture, two short tutorial videos, AI Instructor presentation, two homework sets, and a 40-question assessment.

Mastery standard: 70%+. Students below mastery review the instructional presentation and may retake the assessment, matching the existing LucidPrep secondary model.

U.S. Alignment Note

The U.S. has no single national curriculum, but Math 7 is the standard Grade 7 mathematics course in most districts. This syllabus uses the Common Core State Standards for Mathematics as the primary national reference (adopted by the large majority of states) and cross-checks against California Common Core and Texas TEKS, which cover the same content under different codes.

Prepared for LucidPrep • Curriculum Planning Edition • 2026

Common Instructional & Assessment Model

Weekly Learning Cycle

Step	Component	Description
1	Opening / Domain Overview	~2-minute AI Instructor introduction
2	Objectives + Vocabulary	Student-facing outcomes and core terminology
3	Warm-Up	5 questions with explanations
4	Main Lecture	Concept instruction, models and examples
5	Video Tutorials	2 short instructional videos
6	AI Instructor Presentation	Visual models, examples, strategies and applications
7	Homework	Two sets of 5 questions
8	Weekly Assessment	40 questions; 70% mastery
9	Review / Retake	Below mastery → review presentation → retake

Assessment Philosophy

Assessment is continuous and mastery-based. Each week pairs low-stakes practice (five warm-ups and two homework sets) with a 40-question weekly assessment. Students scoring below 70% review the instructional presentation and retake, so mastery is reached before progressing. Module assessments at Weeks 9, 18 and 27 and a cumulative final at Week 36 measure retention and synthesis.

MATH 7

Grade 7 mathematics aligned to
the Common Core State
Standards

This 36-week course is designed around **the Common Core Grade 7 domains: Ratios & Proportional Relationships (7.RP), The Number System (7.NS), Expressions & Equations (7.EE), Geometry (7.G), and Statistics & Probability (7.SP)**. It is a full-year curriculum with four nine-week modules and a consistent LucidPrep mastery workflow.

Grade	Grade 7
Course	Math 7 · Common Core
Duration	36 weeks
Structure	4 modules × 9 weeks
Weekly assessment	40 questions
Mastery threshold	70%
Module assessments	Weeks 9, 18, 27
Final assessment	Week 36

Course Learning Outcomes

- Analyze proportional relationships and use them to solve real-world problems.
- Operate fluently with rational numbers (integers, fractions and decimals).
- Generate, interpret and solve expressions, equations and inequalities.
- Solve geometry problems involving scale, angle, area, surface area and volume.
- Use sampling and probability to draw inferences and reason about chance.

MATH 7**Module 1 — Rational Numbers & Operations**

Week	Topics	Students will be able to	Standards
1 — Integers & the Number Line	Positive/negative numbers, absolute value, ordering, real-world contexts	Represent and order integers; interpret absolute value	7.NS.1
2 — Adding & Subtracting Rational Numbers	Integer addition/subtraction, additive inverse, distance on number line	Add and subtract rational numbers	7.NS.1
3 — Multiplying & Dividing Rational Numbers	Sign rules, products and quotients, properties	Multiply and divide rational numbers	7.NS.2
4 — Fractions, Decimals & Rational Numbers	Converting forms, terminating/repeating decimals	Convert among forms; classify rational numbers	7.NS.2/3
5 — Operations with Fractions	All four operations with fractions and mixed numbers in context	Solve problems using fraction operations	7.NS.3
6 — Order of Operations & Properties	Expressions with rationals, properties of operations	Evaluate multi-step expressions	7.NS.3; 7.EE.1
7 — Real-World Rational Number Problems	Multi-step problems: temperature, money, elevation	Model and solve real-world problems	7.NS.3
8 — Estimation & Reasoning	Estimating, reasonableness, mental math with rationals	Estimate and assess reasonableness	7.NS.3
9 — Module 1 Assessment	Cumulative review of rational-number operations	Demonstrate mastery of Module 1	7.NS

Module delivery notes. Ground every topic in real-world contexts and multiple representations (tables, graphs, equations, diagrams). Emphasize proportional reasoning and fluency with rational numbers.

Module 2 — Ratios & Proportional Relationships

Week	Topics	Students will be able to	Standards
1 — Ratios & Unit Rates	Ratios, unit rates (including with fractions), rate language	Compute unit rates	7.RP.1
2 — Recognizing Proportional Relationships	Tables, graphs, equivalent ratios, line through origin	Decide whether quantities are proportional	7.RP.2a
3 — Constant of Proportionality	Identify k in tables, graphs, equations, diagrams	Find and interpret the constant of proportionality	7.RP.2b
4 — Representing Proportional Relationships	Equations $y = kx$; meaning of $(0,0)$ and $(1,r)$	Represent proportional relationships by equations	7.RP.2c/d
5 — Percent Problems	Percent as a rate, percent of a number, percent equations	Solve percent problems	7.RP.3
6 — Percent Applications	Tax, tip, markup/markdown, commission, simple interest, percent change	Solve multi-step percent problems	7.RP.3
7 — Scale Drawings	Scale factor, actual lengths and areas from drawings	Solve scale-drawing problems	7.G.1
8 — Proportional Reasoning in Context	Mixed applications, comparing rates	Apply proportional reasoning to new problems	7.RP.3
9 — Module 2 Assessment	Cumulative review of proportional relationships	Demonstrate mastery of Module 2	7.RP

Module delivery notes. Ground every topic in real-world contexts and multiple representations (tables, graphs, equations, diagrams). Emphasize proportional reasoning and fluency with rational numbers.

MATH 7

Module 3 — Expressions, Equations & Inequalities

Week	Topics	Students will be able to	Standards
1 — Algebraic Expressions	Variables, terms, translating verbal to algebraic	Write and interpret expressions	7.EE.1
2 — Equivalent Expressions	Distributive property, combining like terms, factoring	Generate equivalent expressions	7.EE.1
3 — Interpreting Expressions in Context	Rewriting to reveal meaning (increase by 5% = $\times 1.05$)	Relate expressions to real contexts	7.EE.2
4 — Solving Two-Step Equations	$px + q = r$, solving and checking	Solve two-step equations	7.EE.4a
5 — Equation Word Problems	Modeling real problems with equations	Model and solve problems with equations	7.EE.3/4a
6 — Solving Two-Step Inequalities	$px + q > r$, graphing on a number line	Solve and graph inequalities	7.EE.4b
7 — Multi-Step Problems with Rationals	Multi-step real-life problems, estimation	Solve multi-step problems	7.EE.3
8 — Applications & Review	Mixed modeling with expressions and equations	Apply expressions, equations and inequalities	7.EE.3/4
9 — Module 3 Assessment	Cumulative review of expressions and equations	Demonstrate mastery of Module 3	7.EE

Module delivery notes. Ground every topic in real-world contexts and multiple representations (tables, graphs, equations, diagrams). Emphasize proportional reasoning and fluency with rational numbers.

Module 4 — Geometry, Statistics & Probability

Week	Topics	Students will be able to	Standards
1 — Angle Relationships	Complementary, supplementary, vertical, adjacent angles	Use angle relationships to solve problems	7.G.5
2 — Triangles & Constructions	Conditions for triangles, drawing figures	Construct and describe triangles	7.G.2
3 — Circles: Area & Circumference	Pi, area and circumference of circles	Compute area and circumference of circles	7.G.4
4 — Area, Surface Area & Volume	2D/3D figures, prisms, cross-sections	Solve area, surface-area and volume problems	7.G.6/3
5 — Sampling & Inferences	Populations, random and representative samples	Use sampling to draw inferences	7.SP.1/2
6 — Comparing Populations	Comparing two data sets, center and variability	Compare two populations	7.SP.3/4
7 — Probability Concepts	Likelihood, probability 0-1, experimental vs theoretical	Reason about the probability of events	7.SP.5/6
8 — Probability Models & Compound Events	Probability models, simulations, compound events	Find probabilities of compound events	7.SP.7/8
9 — Final Assessment	Cumulative final across all modules	Demonstrate mastery of the full course	7.RP-7.SP

Module delivery notes. Ground every topic in real-world contexts and multiple representations (tables, graphs, equations, diagrams). Emphasize proportional reasoning and fluency with rational numbers.

Assessment & Alignment

Assessment Blueprint

Warm-ups: weekly • Homework: weekly • Weekly assessment: 40 questions • Module assessments: Weeks 9 / 18 / 27 • Final: Week 36 • Below 70%: instructional review and retake.

Standards Alignment

Primary framework	CCSS: 7.RP, 7.NS, 7.EE, 7.G, 7.SP
Mathematical Practices	Make sense of problems and persevere; reason abstractly and quantitatively; construct viable arguments and critique reasoning; model with mathematics; use tools strategically; attend to precision; use structure and regularity.
Recurring Concepts	Multiple representations; equivalence; proportional reasoning; structure; modeling; real-world application.
State mapping	Common Core adopted by the large majority of states; Texas TEKS and other state frameworks cover the same grade-level content under different codes.

Content Production Guidance

Author each week as 5 warm-up items, a concept lecture with worked examples and multiple representations, 2 curated videos, an AI presentation with models and strategies, 2 homework sets (5 items each), and a 40-question bank spanning fluency, application and reasoning. Tag every item to its Common Core code.

Curriculum Basis & Source Notes

This Grade 7 mathematics syllabus was generated from the U.S. Common Core State Standards for Mathematics. Content and pacing reflect the standard Math 7 · Common Core course used across most U.S. districts.

Primary reference	Common Core State Standards for Mathematics — Grade 7 domains and the Standards for Mathematical Practice.
State cross-check	California Common Core State Standards for Mathematics.
State cross-check	Texas Essential Knowledge and Skills (TEKS) for Mathematics.
LucidPrep alignment	Mapped to LucidPrep's 36-week, 4-module secondary structure with 70% mastery.

Recommended next production step. Commission item banks per week (40 assessment items plus warm-ups and homework), each tagged to its Common Core code, then load into the LucidPrep authoring pipeline.